

High Frequency Multiplicative Component GARCH ^{♣*}

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Abstract

This paper proposes a new way of modeling and forecasting intraday returns. We decompose the volatility of high frequency asset returns into components that may be easily interpreted and estimated. The conditional variance is expressed as a product of daily, diurnal and stochastic intraday volatility components. This model is applied to a comprehensive sample consisting of 10-minute returns on more than 2500 US equities.

JEL Classifications: C22, G15

Keywords: Volatility, ARCH, Intra-day Returns.

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1. INTRODUCTION

This paper proposes a new way of modeling and forecasting intraday returns. We decompose the volatility of high frequency asset returns into multiplicative components, which may be easily interpreted and estimated. The conditional variance is expressed as a product of daily, diurnal and stochastic intraday volatility components. This model is applied to a comprehensive sample consisting of 10-minute returns on more than 2500 US equities. We apply a number of different specifications. Namely we build models for separate companies, pool data into industries and consider various criteria for grouping returns. It turns out that results for the pooled regressions seem to be more stable. Therefore it could be expected that the forecasts from the pooled specifications will outperform the corresponding forecasts from company-by-company estimation.

A number of papers have presented related work on intraday returns. The most commonly cited studies include Andersen and Bollerslev (1997, 1998). The authors propose models for 5-minute returns on Deutschemark-dollar exchange rate and the S&P500 index. In the first paper, Andersen and Bollerslev build a multiplicative model of daily and diurnal volatility, and in the second they add an additional component which takes account of macro-economic announcements. For most of their models, the intra-daily volatility components are deterministic. In contrast, the intra-daily components in our model include both deterministic (the diurnal) and stochastic (a separate intra-daily ARCH).

We expect our model to be of particular interest for derivative traders or hedge funds who seek high frequency measures of risk. Another potential use lies in devising the optimal trading strategy in order to reduce both trading costs and the risks of delayed trading. The paper is organized as follows: Section 2 presents the model. Section 3 describes the data and gives results of estimation. Conclusions and an outline of further research are offered in Section 4.

2. THE MODEL

2.1. Notation

We use the following notation. Days in the sample are indexed by t ($t = 1, \dots, T$). Each day is divided into 10 minute intervals referred to as bins and indexed by i ($i = 1, \dots, N$). Current

period is $\{t, i\}$. Price of an asset at day t and bin i is denoted by $P_{\{t, i\}}$. The continuously compounded return $r_{\{t, i\}}$ is modeled as:

$$\begin{aligned} r_{\{t, i\}} &= \ln\left(\frac{P_{\{t, i\}}}{P_{\{t, i-1\}}}\right) & \text{for } i \geq 2 \\ &= \ln\left(\frac{P_{\{t, 1\}}}{P_{\{t-1, N\}}}\right) & \text{for } i = 1 \end{aligned}$$

2.2. Model

We propose a new GARCH model for high frequency intraday financial returns, which specifies the conditional variance to be a multiplicative product of daily, diurnal and stochastic intraday volatility. Intraday equity returns are described by the following process:

$$r_{\{t, i\}} = \sqrt{h_t s_i q_{\{t, i\}}} \varepsilon_{\{t, i\}} \quad \text{and} \quad \varepsilon_{\{t, i\}} \sim N(0, 1) \quad (1)$$

where:

- h_t is the daily volatility component,
- s_i is the diurnal (calendar) volatility pattern,
- $q_{\{t, i\}}$ is the intraday volatility component, and
- $\varepsilon_{\{t, i\}}$ is an error term.

The daily volatility component could be obtained in a number of ways. Similarly to Andersen and Bollerslev (1997, 1998), this component could be estimated from a daily ARCH model for a longer sample, going back a number of months or years. We adopt a different route, however, and utilize commercially available volatility forecasts produced daily for each company in our sample.

The diurnal component is calculated as standard deviation of returns in each bin after deflating by the daily volatility. To see this consider the variance of these returns:

$$\frac{r_{i,t}^2}{h_t} = s_i q_{i,t} \varepsilon_{i,t}^2$$

and

$$E\left(\frac{r_{i,t}^2}{h_t}\right) = s_i E(q_{i,t}) = s_i (\omega / (1 - \alpha - \beta)) \sim s_i$$

Practically, we estimate the model in two stages. First we normalize returns by daily and diurnal volatility components, and then model the residual volatility as a GARCH(1,1) process:

$$y_{\{t,i\}} = r_{\{t,i\}} / \sqrt{\hat{h}_t \hat{s}_i} = \sqrt{q_{\{t,i\}}} \varepsilon_{\{t,i\}} \quad (2)$$

$$q_{\{t,i\}} = \omega + \alpha (r_{\{t,i-1\}} / \sqrt{\hat{h}_t \hat{s}_{i-1}})^2 + \beta q_{\{t,i-1\}} \quad (3)$$

3. EMPIRICAL RESULTS

3.1. DATA

Our sample consists of price data on 2516 companies obtained from the TAQ database. We analyze logarithmic returns standardized by a commercially available volatility forecast for each company and each day and the standard deviation of returns in each 10-minute bin. Data spans a three-month period in April-June 2000.

Some results will be presented using a single randomly chosen NYSE-listed stock – Republic Group Inc. In the following, we will refer to the company by its ticker symbol RGC. We have its daily price data since August 1985 and tick-by-tick data during the period from January 2, 1998 to June 30, 1999. This return series has been previously used by Engle and Patton (2004) and comes from their 4th decile for trading frequency. Over the examined period (1.5 years) it had almost 13 thousand trades or about 35 per day. On the basis of this ultra-high frequency data we calculate 10-minute returns based on closing midquotes, which are used in the subsequent analysis.

3.2. RESULTS FOR A SINGLE STOCK

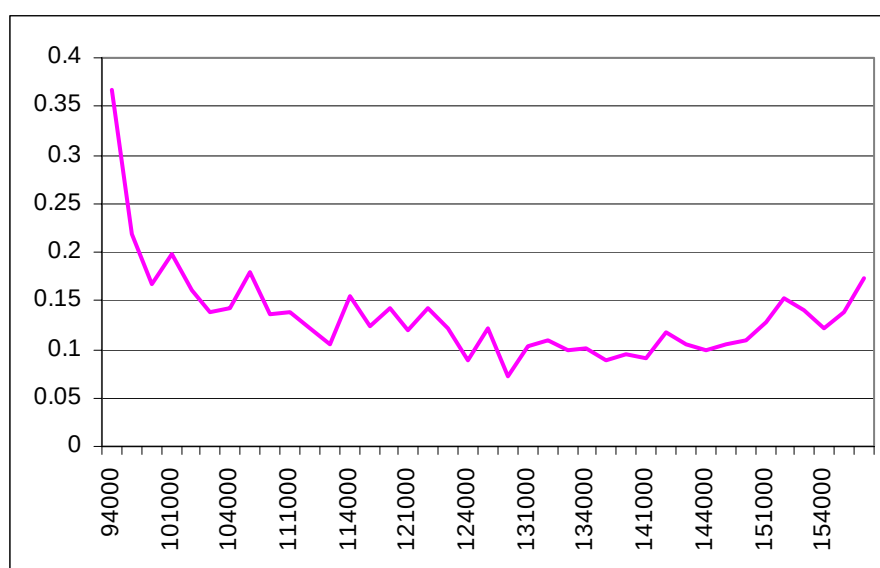
In this section we apply this method to a single stock – RGC. We will refer to the notation of model (2)-(3). We obtain forecasts of daily volatilities h_t from a se-

quence of daily GARCH(1,1) models estimated from August 1985 until one day before the forecast period. This procedure is similar to the daily volatility component estimation applied by Andersen and Bollerslev (1997, 1998). Daily forecasts are computed for RGC between January 2, 1998 and June 30, 1999. Next we divide 10-minute returns by their respective daily volatility forecasts. What can be observed for these data, however, is a very pronounced diurnal volatility pattern. Figure 1 plots the standard deviation of returns in each of 40 10-minute bins. There is a pronounced increased variation in the beginning of each day, a calm period in the middle and somewhat increased variation towards the end. This diurnal pattern has been observed by many studies for all sorts of financial returns.

The sample standard deviation for each bin will be our estimate of diurnal component s_i . Hence in the second step, returns are normalized by their respective diurnal standard deviations. In order to take account of the remaining intraday dynamics, we fit a GARCH (1,1) model into returns standardized in that way. Tables 1 and 2 present results of daily and intraday GARCH estimation. Figure 2 superimposes the three volatility components described above. We have chosen to only show two-week period from our sample for the reason of clarity.

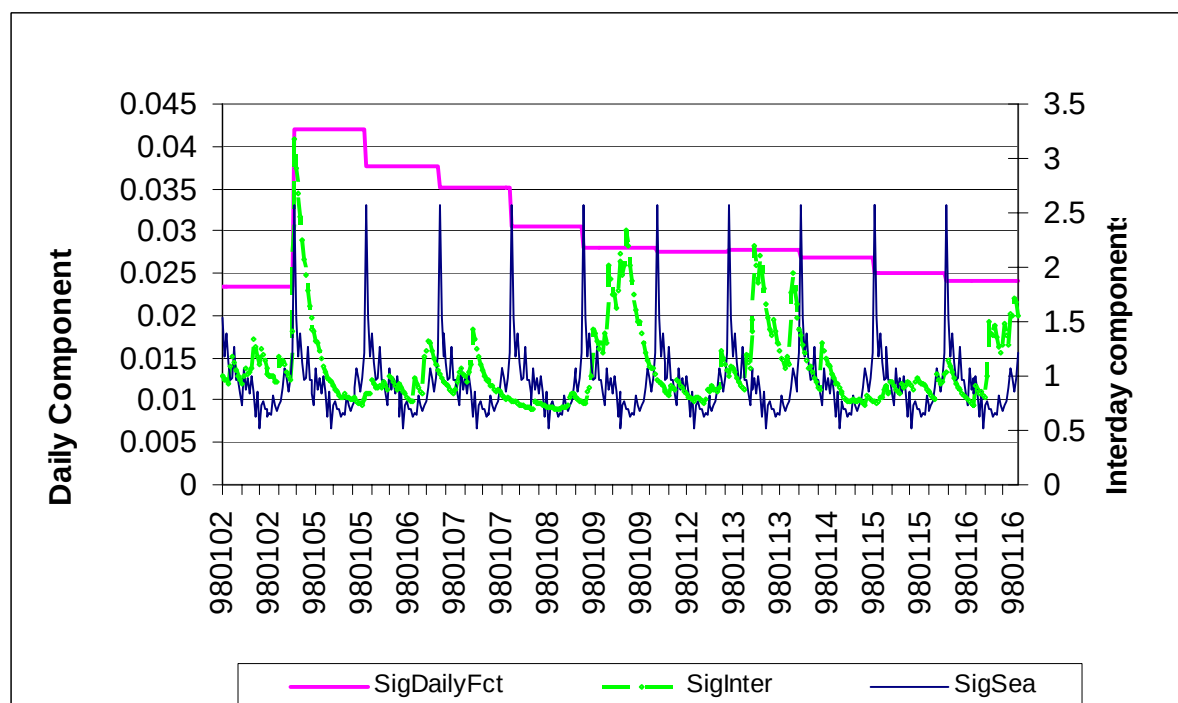
The pink rectangular line denoted as “SigDailyFct” shows daily volatility forecast, common for each day. Navy-blue “SigSea” represents the regular diurnal pattern, and the stochastic intraday component in green dashed line SigInter depicts the intraday volatility component. We may appreciate that this component is able to modify the regular deterministic diurnal pattern.

Figure 1
Standard deviation of return during a day for the RGC Stock



Note: The horizontal axis labels denote hours during a trading day.

Figure 2
Volatility Components for the RGC Stock



Note: The horizontal axis indicates days.

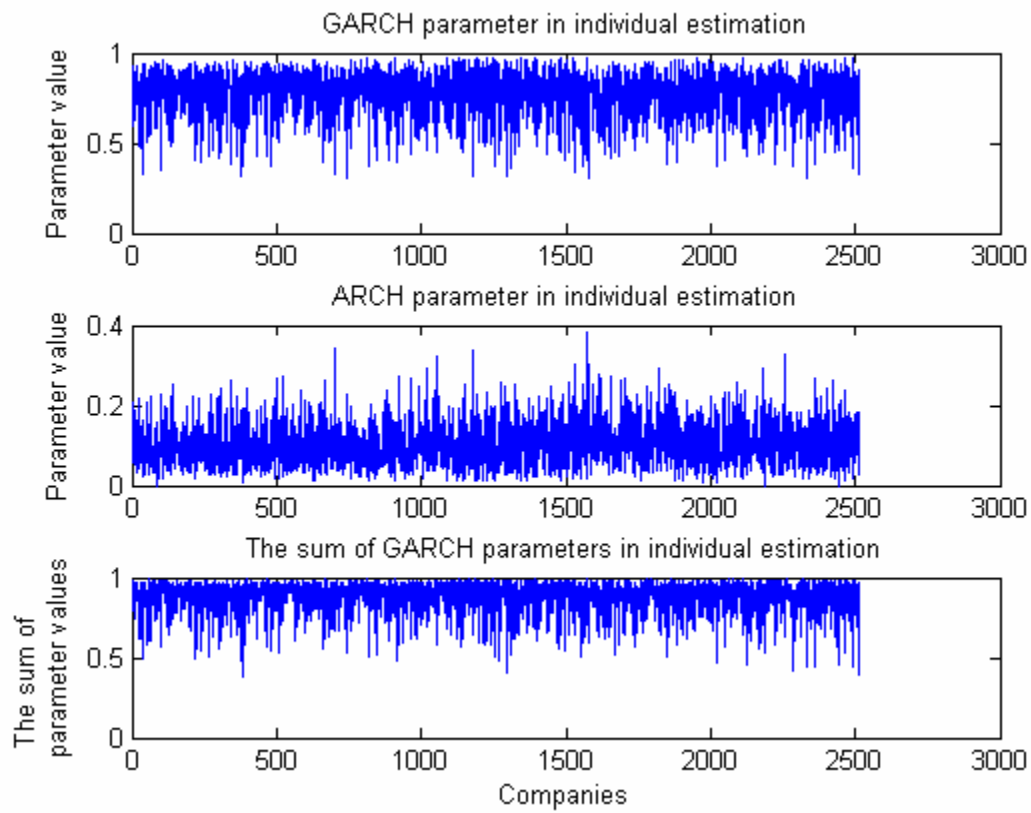
3.3. RESULTS FOR A SAMPLE OF 2516 STOCKS

3.3.1 SEPARATE ESTIMATION RESULTS

Model (2)-(3) is estimated for equity returns, which have been previously divided by a volatility forecast for a day and “diurnally adjusted” by the standard deviation for each bin. As mentioned before we use a comprehensive sample of 2516 US stocks. Estimation is performed for the period April-May 2000, and the combined count of observations during this period exceeds 4.2 million data points. Since it is rather demanding to fit results of this estimation into a table of a manageable size, we report results of this procedure resorting to graphical methods. By a “GARCH parameter” and an “ARCH parameter”, we refer to β and α coefficients from equation (3). The top and middle panels of Figure 3 depict β and α parameters, respectively. The bottom panel plots the sum of both parameters, thus informing us how persistent the volatility is. We may observe a fair amount of variation in the values of parameters and the measure of persistence. This result cannot be surprising taking into account that the estimation sample spans only two months. Although for 10-minute returns this translates into 1560 observations for most companies, the actual interval appears to be very short. Figure 4 sheds some more light on the nature of the observed parameter variation in separate regressions. It reports results of the very same estimations; however the sequence of companies has changed. We have sorted the companies according to the percentage of “zero” returns. This percentage could roughly be understood as a measure of intensity of trading. Substantial percentage of zero returns would indicate lack of active trading (although a zero return may result from trading at the same price).

Companies at the very left of Figure 4 almost always trade, and at the very right hardly ever trade or trade at the same price most of the time. It can be observed that estimates’ variability increases with the percentage of zero returns. Further, there is an upward trend in the GARCH parameter and a downward tendency for the ARCH parameter. The latter one is more noticeable in Figure 4, due to a different scale of the middle panel.

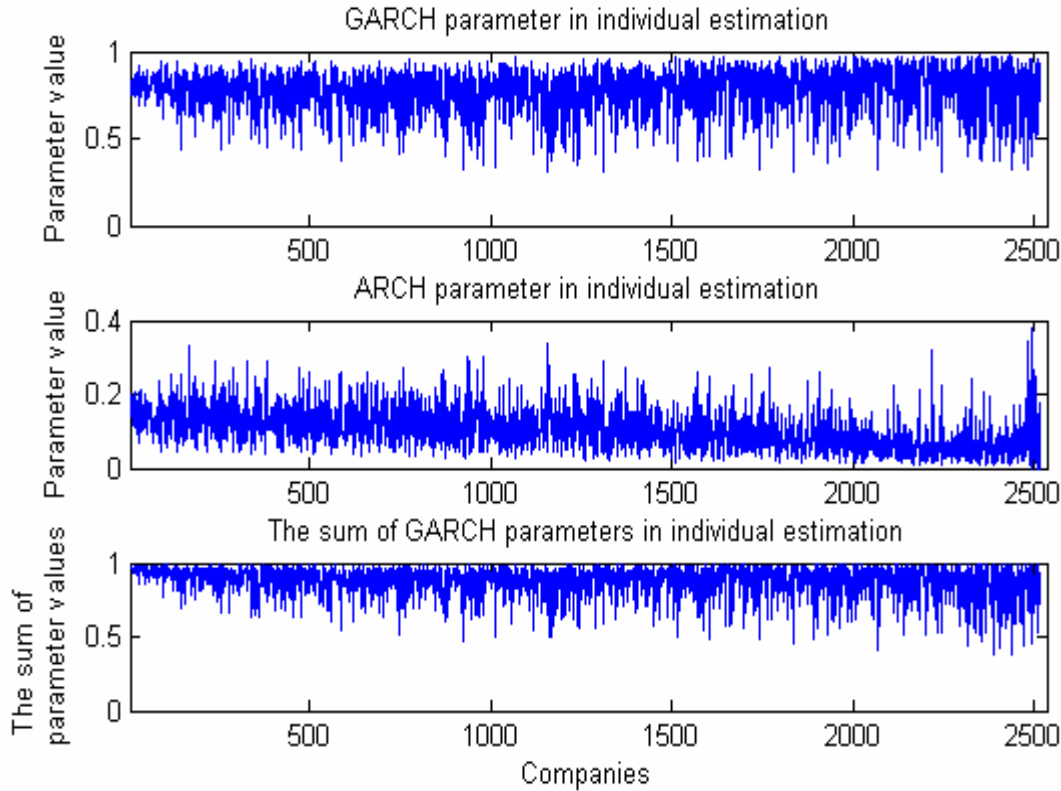
Figure 3
GARCH Estimation Results for the Intra-Day Component
2516 Separate Models



Note: The horizontal axis indicates companies.

Figure 4

GARCH Estimation Results for the Intra-Day Component
2516 Separate Models, Companies Sorted According to Their Trading Intensity



Note: The horizontal axis indicates particular companies.

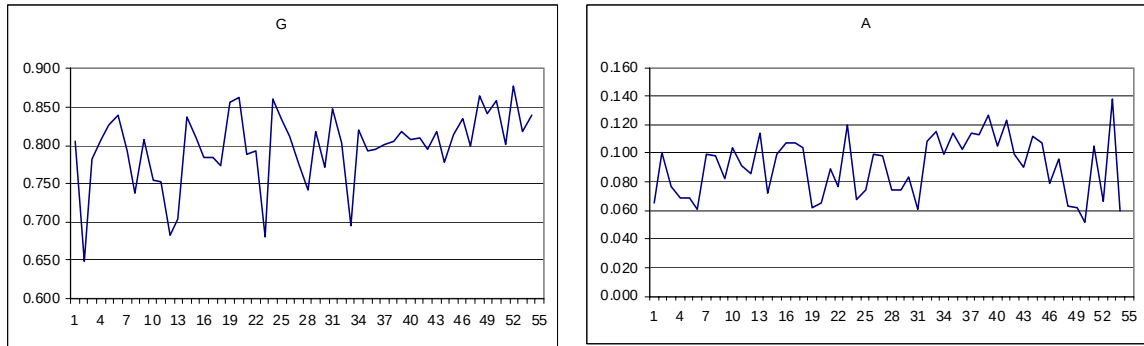
3.3.2 GROUPED ESTIMATION RESULTS

In order to obtain superior forecasting performance of our model, we have decided to group data into 54 industries and estimate 54, instead of 2516 GARCH (1,1) models for an intraday component. Each return series has been divided by its standard deviation in order to render returns comparable across stocks. Standardized returns within each industry have been pooled into industry series. Estimation results of this step are summarized in Table 3 and parameters plotted in Figure 5. The last column indicates a very substantial reduction in persistence of volatility.

One concern here is that industries vary in number of companies. We are currently investigating ways to group some of the smaller industries into bigger structures.

Table 4 reports results for ARCH estimation for the intraday component for one giant pool of all the companies, comprising over 4.2 million observations. Similarly to the industry case, this table also indicates modest persistence of intraday volatility.

Figure 5
GARCH Estimation Results for the Intra-Day Component
54 Industry Models



Note: G and A denote β and α parameters in model (2)-(3), respectively.

4. CONCLUSION

This paper proposes a new way of modeling and forecasting intraday returns. We decompose the volatility of high frequency asset returns into components that may be easily interpreted and estimated. The conditional variance is expressed as a product of daily, diurnal and stochastic intraday volatility components. This model is applied to a comprehensive sample consisting of 10-minute returns on more than 2500 US equities.

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Table 1. RGC Daily GARCH Results

Parameter	Value	Standard Error	T Statistic
-----	-----	-----	-----
C	0.00023757	0.00041135	0.577537
K	0.00013636	0.000011494	11.86358
GARCH (1)	0.65957	0.019085	34.5596
ARCH (1)	0.18073	0.010675	16.93021

Note: C denotes a constant in the mean equation, and K - a constant in the variance equation

Table 2. RGC Intraday GARCH Results

Parameter	Value	Standard Error	T Statistic
-----	-----	-----	-----
C	-0.0047551	0.0072439	-0.6564
K	0.079635	0.0020221	39.3828
GARCH (1)	0.8311	0.0033093	251.1402
ARCH (1)	0.097794	0.002116	46.2155

Note: C denotes a constant in the mean equation, and K - a constant in the variance equation

Table 3. Industry Results (Notes on the next page)

Industry	NoObs	LLF	C	tC	K	tK	G	tG	A	tA	BIC	1	2	5	10	20	40	A+G	sum
1	92976	-128890	-0.0140	-4.61	0.132	121.48	0.805	558.34	0.065	110.66	257820	1.993	4.76	7.58	11.68	33.56	71.31	0.870	
2	24102	-33598	-0.0110	-1.83	0.256	61.22	0.649	120.20	0.101	54.37	67236	0.011	0.77	1.45	2.39	3.82	13.97	0.750	
3	71292	-98840	-0.0155	-4.49	0.144	134.95	0.783	533.04	0.076	92.81	197720	0.423	1.34	5.13	10.66	73.71	129.84	0.859	
4	134200	-186330	-0.0074	-2.90	0.125	120.58	0.808	557.18	0.069	107.76	372700	1.930	3.74	7.80	12.97	24.02	161.73	0.877	
5	100580	-139360	0.0116	3.95	0.107	118.80	0.826	649.19	0.069	98.56	278770	0.458	1.01	6.61	13.20	103.65	147.08	0.895	
6	43641	-60851	-0.0017	-0.38	0.101	52.28	0.840	317.81	0.061	52.63	121740	0.643	2.22	3.66	5.13	10.05	39.47	0.901	
7	55380	-76573	0.0034	0.86	0.111	64.10	0.793	292.88	0.100	62.23	153190	0.999	1.03	43.51	44.50	65.38	86.19	0.893	
8	102610	-141890	0.0029	0.99	0.169	110.13	0.737	336.33	0.098	95.58	283830	0.359	2.72	4.46	15.16	30.18	69.32	0.834	
9	23673	-32638	0.0101	1.74	0.118	78.44	0.808	358.65	0.082	66.39	65317	0.238	0.27	2.07	4.25	7.86	118.25	0.890	
10	10647	-14695	0.0037	0.44	0.150	48.29	0.755	178.30	0.104	36.45	29426	0.445	0.45	1.45	7.01	142.95	160.76	0.859	
11	37323	-51825	0.0050	1.04	0.159	68.97	0.752	254.07	0.092	81.17	103690	0.442	1.07	2.71	15.79	22.34	76.92	0.844	
12	28236	-39090	-0.0024	-0.42	0.232	62.43	0.683	140.03	0.086	45.27	78222	0.001	0.20	1.55	3.39	7.48	253.32	0.769	
13	33540	-46304	-0.0157	-3.19	0.187	54.69	0.703	133.14	0.114	43.30	92651	0.015	0.25	0.96	2.18	3.79	6.34	0.817	
14	56978	-78622	-0.0208	-5.58	0.094	90.88	0.837	526.71	0.072	79.95	157290	2.048	2.08	9.04	11.63	28.75	57.13	0.908	
15	44928	-60880	-0.0081	-1.94	0.094	90.89	0.809	449.96	0.100	80.36	121800	3.571	3.61	11.13	15.59	23.61	65.68	0.909	
16	37401	-51323	0.0002	0.05	0.112	57.39	0.785	237.54	0.107	55.84	102690	1.898	2.61	6.81	12.28	36.76	47.13	0.892	
17	90714	-123720	-0.0072	-2.42	0.113	146.53	0.783	562.79	0.107	100.38	247480	0.450	1.21	4.20	6.62	52.08	100.34	0.891	
18	24960	-34412	-0.0038	-0.67	0.125	62.90	0.774	257.30	0.104	52.14	68864	1.693	2.48	7.71	14.57	24.90	42.96	0.878	
19	84240	-115920	-0.0061	-1.95	0.083	128.23	0.856	830.67	0.062	113.08	231880	1.461	1.50	2.83	7.67	11.90	28.08	0.918	
20	57330	-79145	-0.0151	-3.93	0.074	81.01	0.863	587.80	0.066	74.98	158330	2.166	2.17	9.64	20.53	26.12	43.01	0.928	
21	103230	-141720	-0.0137	-4.72	0.126	153.10	0.788	601.23	0.089	104.86	283480	0.014	0.37	1.81	4.52	8.36	22.54	0.877	
22	28119	-39152	0.0030	0.54	0.134	41.77	0.792	184.19	0.076	42.78	78346	0.690	1.00	5.21	7.52	16.13	51.38	0.869	
23	37401	-51842	-0.0024	-0.51	0.206	50.80	0.680	129.13	0.120	60.64	103730	0.914	1.97	9.92	12.01	39.23	55.49	0.799	
24	25584	-34907	-0.0061	-1.09	0.073	58.53	0.860	438.01	0.068	55.61	69854	0.473	0.50	8.97	10.34	25.12	75.87	0.928	
25	48165	-66239	-0.0071	-1.69	0.095	92.02	0.832	529.08	0.075	71.41	132520	3.651	3.69	6.60	13.66	39.91	65.51	0.907	
26	10920	-14976	-0.0062	-0.73	0.091	19.69	0.811	114.01	0.099	24.81	29988	3.649	3.69	5.12	12.46	19.14	48.62	0.910	
27	42119	-57852	-0.0084	-1.87	0.127	63.59	0.777	237.50	0.099	55.96	115750	0.233	1.42	1.89	4.56	8.10	646.70	0.875	
28	17121	-23836	0.0027	0.37	0.185	32.55	0.742	102.24	0.074	31.63	47711	0.092	0.88	0.99	2.34	7.47	32.70	0.816	
29	103540	-142960	-0.0063	-2.18	0.112	142.93	0.817	689.12	0.074	116.15	285970	0.303	1.82	4.61	9.90	13.52	24.18	0.891	
30	119110	-166030	0.0113	4.12	0.147	138.16	0.771	522.60	0.083	93.20	332100	5.716	6.45	23.68	28.00	118.85	153.48	0.854	
31	58733	-81804	0.0121	3.16	0.093	76.81	0.848	493.08	0.061	71.20	163650	1.023	1.26	4.17	10.34	26.81	114.25	0.909	
32	15600	-21405	0.0010	0.14	0.094	25.74	0.803	135.94	0.109	30.64	42850	0.055	1.09	3.27	8.50	18.50	34.50	0.911	
33	31629	-43650	-0.0036	-0.70	0.196	78.15	0.695	194.59	0.116	58.08	87342	0.031	1.14	2.64	5.73	26.91	66.64	0.811	
34	44694	-60711	-0.0031	-0.79	0.088	153.28	0.820	868.30	0.100	109.77	121470	2.506	3.29	5.75	8.59	12.75	74.63	0.920	
35	51401	-70025	0.0006	0.14	0.097	63.02	0.794	329.90	0.114	76.98	140090	14.129	15.06	20.17	25.39	40.37	72.11	0.908	
36	108810	-148910	-0.0043	-1.65	0.108	127.52	0.795	606.19	0.102	124.64	297860	4.554	5.46	11.45	18.69	22.48	45.36	0.897	
37	214610	-290520	-0.0122	-6.57	0.089	161.37	0.801	891.12	0.114	188.33	581080	3.368	3.55	8.31	16.64	80.34	109.06	0.916	
38	222300	-298840	-0.0130	-7.17	0.084	191.53	0.806	952.12	0.113	169.34	597720	4.439	7.57	21.20	44.04	50.19	100.56	0.919	
39	148820	-197780	-0.0146	-6.82	0.061	108.20	0.818	744.25	0.126	131.09	395600	16.126	17.58	45.50	73.92	81.46	193.38	0.944	
40	125260	-168830	-0.0178	-7.20	0.088	174.91	0.807	801.51	0.105	132.50	337710	0.368	0.52	3.31	5.79	8.46	632.56	0.913	
41	258060	-344260	-0.0236	-14.61	0.073	228.02	0.810	1372.00	0.123	222.82	688570	16.312	17.97	39.25	60.83	68.41	160.80	0.932	
42	34242	-46826	0.0016	0.32	0.109	60.39	0.795	282.90	0.100	59.50	93694	3.999	4.33	9.42	12.43	21.39	57.09	0.895	
43	113020	-153890	-0.0180	-6.84	0.092	129.93	0.819	662.60	0.091	109.66	307820	0.926	2.41	10.34	18.61	24.77	127.73	0.909	
44	50505	-68427	-0.0307	-7.79	0.116	123.27	0.777	467.03	0.112	86.16	136900	0.021	0.08	0.97	3.90	12.98	72.66	0.889	
45	174680	-235800	-0.0157	-7.75	0.085	184.04	0.814	935.66	0.107	154.84	471660	4.365	5.84	16.64	26.94	32.91	66.18	0.920	
46	85602	-117200	-0.0031	-1.04	0.089	118.16	0.835	729.52	0.079	116.18	234450	5.362	5.46	15.80	20.04	40.39	58.82	0.914	
47	34865	-47537	0.0058	1.21	0.109	89.12	0.799	429.74	0.096	84.32	95115	5.988	6.95	12.86	17.93	27.10	62.16	0.895	
48	96524	-132830	0.0049	1.72	0.074	169.03	0.865	1506.90	0.063	128.44	265700	14.657	14.72	17.65	24.20	30.03	212.02	0.928	
49	218210	-302410	0.0067	3.48	0.098	185.07	0.842	1064.50	0.062	149.24	604880	3.776	4.97	10.41	16.23	26.80	154.44	0.904	
50	48048	-66832	0.0039	0.95	0.090	74.70	0.859	533.63	0.052	71.95	133710	2.984	4.15	7.04	10.86	12.75	36.55	0.911	
51	57679	-78235	-0.0029	-0.79	0.097	117.88	0.802	535.37	0.105	89.48	156510	0.370	0.85	4.39	8.74	15.65	41.16	0.907	
52	87903	-119860	-0.0061	-2.02	0.060	156.60	0.876	1391.00	0.066	131.10	239760	4.911	4.99	6.84	9.02	20.79	98.63	0.943	
53	79754	-104390	-0.0284	-10.46	0.050	74.06	0.817	501.03	0.137	97.43	208830	10.193	10.20	23.44	42.19	50.96	120.91	0.955	
54	156160	-218270	0.0133	5.73	0.104	97.56	0.838	575.22	0.059	101.57	436580	1.455	2.40	7.01	17.43	45.70	64.92	0.898	
ARCH test critical values												3.842	5.99	11.07	18.31	31.41	55.76		

Notes to Table 3

NoObs	Number of observations
LLF	Value of the Loglikelihood function
C	Constant in the mean equation
tC	t-value for C parameter
K	Constant in the variance equation
tK	t-value for K parameter
G	beta in the variance equation
tG	t-value for G
A	alpha in the variance equation
tA	t-value for A
BIC	Schwartz Information Criterion
1	ARCH LM test for volatility clustering in residuals – Lag 1
2	Lag 2
5	Lag 5
10	Lag 10
20	Lag 20
40	Lag 40
A+G	the sum of alpha and beta parameters

Table 4 All Sample Results

Parameter	Value	Standard Error	T Statistic
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C	-0.0072459	0.00043311	-16.7301
K	0.099443	0.00011118	894.4290
GARCH (1)	0.81556	0.00018085	4509.5115
ARCH (1)	0.087421	0.00011767	742.9462

Note: C denotes a constant in the mean equation, and K - a constant in the variance equation